

Sub-Sec	Subject	Qs	Marks
b-1	Elements of Electrical Engg.	9	9
b-2	Elements of Electronics Engg.	8	8
b-3	Elements of Computer Engg.	8	8
TOTAL		25	25

b-1 ELEMENTS OF ELECTRICAL ENGINEERING (9 Questions | 9 Marks)

■ Electric & Magnetic Circuits

- EMF, Ohm's Law $V=IR$; KCL: $\sum I=0$ at node; KVL: $\sum V=0$ in loop
- Faraday's Laws: $e=-d\Phi/dt$; $e=Blv$; Lenz's Law
- Network Theorems: Thevenin, Norton, Superposition, Maximum Power Transfer

■ AC Circuits

- RMS: $V_{rms} = V_{peak}/\sqrt{2}$; Average = $0.637 \times V_{peak}$
- Impedance: $Z_R=R$, $Z_L=j\omega L$, $Z_C=1/j\omega C$; Series RLC: $Z=\sqrt{R^2+(X_L-X_C)^2}$
- Resonance: $f_r=1/(2\pi\sqrt{LC})$; at resonance $Z=R$ (series), $Z=\max$ (parallel)
- Power: $P=VI\cos\phi$ (active), $Q=VI\sin\phi$ (reactive), $S=VI$ (apparent)

■ Transformers

- Turns ratio: $N_1/N_2 = V_1/V_2 = I_2/I_1$
- Single-phase & three-phase transformers; Star-Delta connections
- Losses: copper loss (I^2R), iron/core loss (hysteresis + eddy current)

■ DC Machines

- Generator: EMF $E=P\Phi N_z/(60A)$; types — series, shunt, compound
- Motor: Back EMF $E_b=V-I_aR_a$; torque $T\propto\Phi I_a$; speed $N\propto E_b/\Phi$
- Speed control methods; starting methods

■ 3-Phase Induction Motor

- $N_s=120f/P$ (synchronous speed); slip $s=(N_s-N_r)/N_s$
- Rotating magnetic field principle; squirrel-cage vs slip-ring
- Torque $\propto sE^2R_2/(R_2^2+s^2X_2^2)$; starting: DOL, star-delta, auto-transformer

■ Transmission & Distribution

- High voltage transmission $\rightarrow$ lower $I \rightarrow$ lower I^2R losses
- 3-phase (4-wire) vs 1-phase; 3-phase advantages: less conductor material
- DC 3-wire vs AC systems comparison

b-2 ELEMENTS OF ELECTRONICS ENGINEERING (8 Questions | 8 Marks)

■ Measurements & Instrumentation

- Errors: absolute, relative, gross, systematic, random; Accuracy vs Precision

- Instruments: Ammeter (series), Voltmeter (parallel), Wattmeter, Energy meter
- CRO: measure V, I, f, phase; Lissajous figures
- Resistance measurement: Wheatstone bridge (medium), Kelvin (low), Megger (insulation)
- Transducers: thermocouple (temp), LVDT (displacement), strain gauge

■ Analog Electronics

- Diode: forward/reverse bias; half-wave/full-wave rectifier; filter circuits
- Zener diode: breakdown voltage regulator; LED, photodiode working
- BJT: NPN/PNP; CB, CE, CC configs; biasing; α , β , γ relations
- FET: JFET & MOSFET; characteristics; transconductance g_m
- Amplifiers: small signal (CE amp) vs large signal (power amp classes A/B/AB/C)
- OP-AMP: inverting/non-inverting; virtual ground; applications: adder, differentiator, integrator, comparator
- Oscillators: Hartley, Colpitts, RC phase shift, crystal; Barkhausen criterion
- Regulated Power Supply: Zener/transistor regulator; IC 78xx/79xx series

■ Digital Electronics

- Number systems: Binary, Octal, Hex; conversions BCD, Gray code; parity bit
- Binary arithmetic: addition, subtraction (1's/2's complement)
- Logic gates: AND, OR, NOT, NAND, NOR, XOR, XNOR; truth tables; symbols
- Boolean algebra: De Morgan's theorems; SOP, POS; K-Map simplification
- Flip-Flops: SR, JK, D, T; clocked; master-slave; registers & counters

■ Industrial Electronics

- SCR (Thyristor): structure, triggering, I-V characteristics; applications in rectifiers
- DIAC: bidirectional trigger diode; used with TRIAC
- TRIAC: bidirectional SCR; AC power control; applications

b-3 ELEMENTS OF COMPUTER ENGINEERING (8 Questions | 8 Marks)

■ Fundamentals of Computers

- Organisation: Input → CPU (ALU + CU + Registers) → Output + Storage
- Data Processing; High-level langs; Translators: Compiler (full translation), Interpreter (line-by-line), Assembler
- Algorithm: step-by-step; Flowchart: symbols (Start/Stop=oval, Process=rect, Decision=diamond, I/O=parallelogram)

■ Computer Organisation

- Registers: MAR, MDR, PC, IR, ACC; Bus: data, address, control
- Instruction Set: RISC vs CISC; fetch-decode-execute cycle
- Hardwired vs Microprogrammed control; Memory hierarchy
- Cache Memory: faster, costly, closer to CPU; Virtual Memory: uses HDD as RAM extension
- I/O: programmed I/O, interrupt-driven, DMA

■ Operating Systems

- Functions: process management, memory management, file system, I/O, security
- Process states: new→ready→running→waiting→terminated
- Scheduling: FCFS, SJF, Round Robin, Priority
- Critical Section: mutual exclusion; Semaphores; Monitors; Deadlock conditions

■ C Programming

- Data types: int, float, char, double; type qualifiers; storage classes
- Console I/O: printf(), scanf(); format specifiers %d %f %c %s
- Control: if-else, switch, for, while, do-while, break, continue
- Arrays: 1D/2D; Strings: string.h functions (strlen, strcpy, strcat, strcmp)
- Structures & Unions; Functions: call by value/reference; recursion
- Pointers: ptr=&var; *ptr; pointer arithmetic; pointers with arrays/functions
- File handling: fopen, fclose, fprintf, fscanf, fread, fwrite modes r/w/a

■ Computer Networking

- LAN (local), WAN (wide area), MAN; Topologies: bus, star, ring, mesh
- Internet: IP address, DNS, HTTP, FTP, SMTP, TCP/IP model vs OSI model
- Applications: email, WWW, FTP, Telnet; bandwidth, latency